IGCSE 0607 Practice Questions — E6.2

Area and perimeter

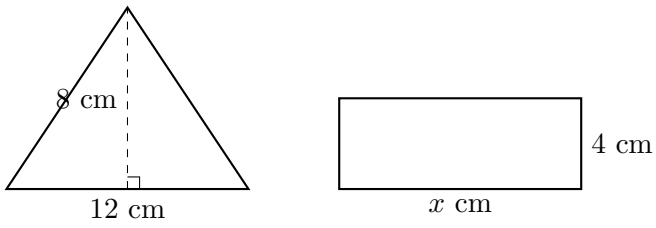
Total Questions: 9 — **Total Marks:** 60 Marks

Table of Contents & Quick Navigation

- **E6.2** Area and perimeter
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

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1 An isosceles triangle and a rectangle are shown below.
The area of the triangle is equal to the area of the rectangle.



NOT TO SCALE

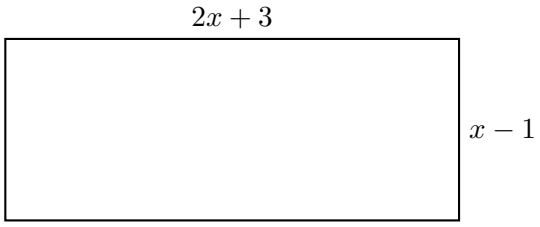
(a) Calculate the value of x.

$x =$ ..... **[2]**

(b) Calculate the perimeter of the isosceles triangle.

..... cm **[3]**

2 The diagram shows a rectangle.
All measurements are in centimetres.



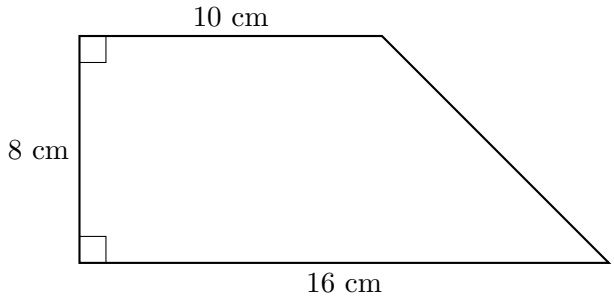
(a) The perimeter of the rectangle is 34 cm.
Find the value of x.

$x =$ ..... **[3]**

(b) Calculate the area of the rectangle.

..... cm^2 **[2]**

3 The diagram shows a right-angled trapezium.



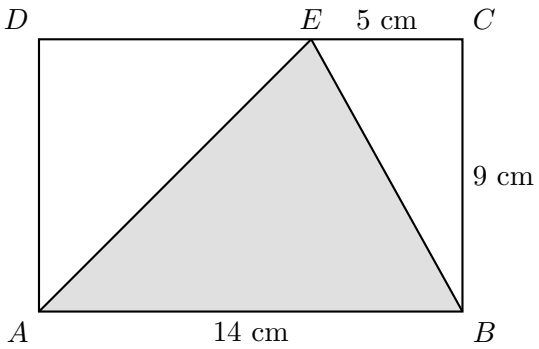
(a) Calculate the area of the trapezium.

..... cm^2 **[2]**

(b) Calculate the perimeter of the trapezium.

..... cm **[3]**

4 $ABCD$ is a rectangle with length 14 cm and width 9 cm.
The point E lies on the edge CD such that $CE = 5$ cm.



NOT TO SCALE

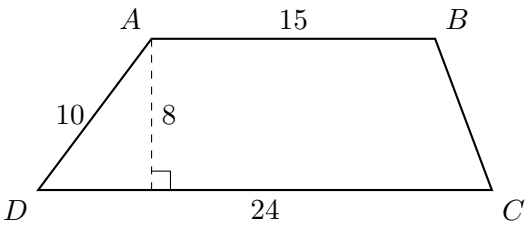
(a) Calculate the area of the shaded triangle ABE.

..... cm^2 **[2]**

(b) Calculate the perimeter of the shaded triangle ABE.

..... cm **[4]**

5 The diagram shows a trapezium $ABCD$ with AB parallel to DC.
The perpendicular height of the trapezium is 8 cm.
All measurements are in centimetres.



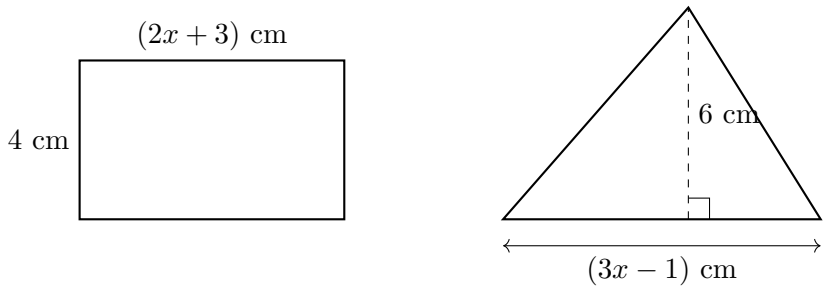
(a) Calculate the area of the trapezium.

..... cm^2 **[2]**

(b) Calculate the perimeter of the trapezium.

..... cm **[4]**

6 The diagram shows a rectangle and a triangle.
The area of the rectangle is equal to the area of the triangle.



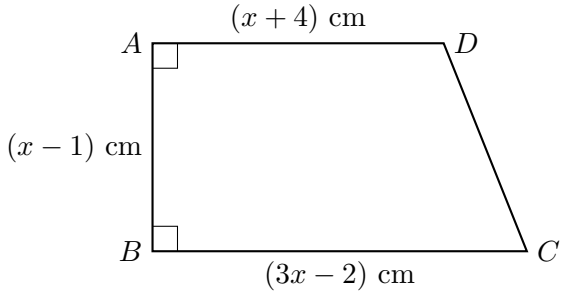
(a) Form an equation in x and solve it to find the value of x.

$x =$ ..... **[4]**

(b) Work out the perimeter of the rectangle.

..... cm **[2]**

7 A trapezium $ABCD$ has right angles at A and B.
The side AD is parallel to BC.
$AD = (x + 4)$ cm, $BC = (3x - 2)$ cm and $AB = (x - 1)$ cm.



NOT TO SCALE

(a) The area of the trapezium is 20 cm^2.

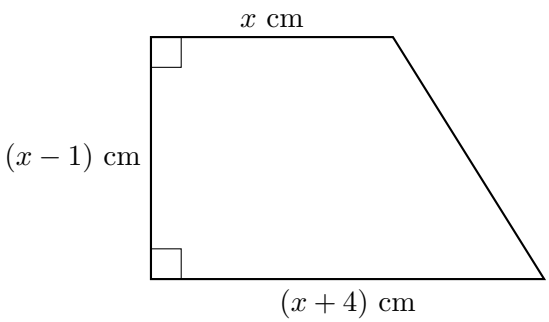
(i) Show that $2x^2 - x - 21 = 0$.

[3]

(ii) Calculate the perimeter of the trapezium.

..... cm **[5]**

8 The diagram shows a right-angled trapezium with parallel sides of length x cm and $(x + 4)$ cm.
The height of the trapezium is $(x - 1)$ cm.
The area of the trapezium is 40 cm^2.



(a) Show that $x^2 + x - 42 = 0$.

[3]

(b) Solve the equation $x^2 + x - 42 = 0$ to find the value of x.

$x =$ ..... **[2]**

(c) Calculate the perimeter of the trapezium.

..... cm **[4]**

9 A piece of wire of length 60 cm is cut into two pieces.
One piece is bent to form a square.
The other piece is bent to form a rectangle where the length is exactly twice the width.
Let the width of the rectangle be y cm.

(a) Write down an expression, in terms of y, for the side length of the square.

..... cm **[2]**

(b) The total area enclosed by the square and the rectangle is 113 cm^2.
Show that $17y^2 - 180y + 448 = 0$.

[3]

(c) Solve the equation.

$$17y^2 - 180y + 448 = 0$$

Show all your working.

$y =$ ..... , $y =$ ..... **[3]**

(d) Given that the side length of the square must be greater than the width of the rectangle, calculate the area of the square.

..... cm^2 **[2]**